

Space Shooter — Complete Build Instructions

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Prerequisites

Requirement	Version
Unity Editor	2021.3 LTS or newer (2022.3 LTS recommended)
Build Support	Windows Build Support (IL2CPP or Mono)
OS	Windows 10/11 (for building .exe)
.NET	.NET Framework 4.x or .NET Standard 2.1

Install Unity

1. Download **Unity Hub** from <https://unity.com/download>
 2. In Unity Hub, install **Unity 2022.3 LTS** (or latest LTS)
 3. During install, ensure **Windows Build Support (Mono)** is checked
 4. Also ensure **2D Template** packages are included
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Project Setup

Option A: Open This Project Directly

1. Open **Unity Hub**
2. Click **Open → Add project from disk**
3. Navigate to this `space_shooter_game` folder and select it

4. Unity will import all scripts and create the necessary `.meta` files
5. Wait for compilation to complete (check bottom status bar)

Option B: Create Fresh Project and Import

1. In Unity Hub, click **New Project**
2. Select the **2D (Built-in Render Pipeline)** template
3. Name it `SpaceShooter` and choose a location
4. Once the empty project opens, close it
5. **Copy these folders** from this project into your new project:
 - `Assets/Scripts/` → entire folder
 - `Assets/Sprites/` → entire folder (if pre-generated)
 - `ProjectSettings/TagManager.asset` → replace existing
 - `ProjectSettings/InputManager.asset` → replace existing
 - `ProjectSettings/Physics2DSettings.asset` → replace existing
6. Reopen the project in Unity

Generate Sprites & Assets

After the project compiles successfully:

1. In Unity, go to the top menu: **Tools → Space Shooter → Generate All Sprites**
2. This will procedurally create all game sprites as PNG files:
 - `Assets/Sprites/Player/player_ship.png`
 - `Assets/Sprites/Enemies/basic_enemy.png`, `fast_enemy.png`, `tank_enemy.png`, `boss_enemy.png`
 - `Assets/Sprites/Bullets/player_bullet.png`, `enemy_bullet.png`, `boss_bullet.png`
 - `Assets/Sprites/PowerUps/powerup_health.png`, `powerup_shield.png`, etc.
 - `Assets/Sprites/Effects/explosion.png`, `shield.png`
 - `Assets/Sprites/Backgrounds/bg_far.png`, `bg_mid.png`, `bg_near.png`
 - `Assets/Sprites/UI/life_icon.png`, `health_fill.png`, `button_bg.png`
3. Wait for Unity to reimport assets (progress bar at bottom)

Configure Sprite Import Settings

After generation, select all sprites in the Project window and verify:

- **Texture Type:** Sprite (2D and UI)
- **Pixels Per Unit:** 64
- **Filter Mode:** Point (no filter) — for pixel-art look
- **Compression:** None

Automated Scene & Prefab Setup

The **Game Setup Wizard** automates everything:

1. Go to menu: **Tools → Space Shooter → Full Game Setup**
2. Click **“Yes, Set Up Everything”** in the confirmation dialog
3. The wizard will automatically:
 - ☒ Generate all sprites (if not already done)

- ☒ Create all prefabs with correct components
 - ☒ Build the **MainMenu** scene with UI
 - ☒ Build the **GamePlay** scene with player, spawner, HUD
 - ☒ Build the **GameOver** scene with UI
 - ☒ Configure Build Settings with all 3 scenes
4. Review the console log for any warnings
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Manual Setup (Alternative)

If the wizard has issues, follow these manual steps:

Create Tags

1. **Edit → Project Settings → Tags and Layers**
2. Add these tags (some may already exist):
 - Player
 - Enemy
 - PlayerBullet
 - EnemyBullet
 - PowerUp

Create Prefabs Manually

Player Prefab

1. Create empty GameObject, name it "Player"
2. Add components:
 - **SpriteRenderer** → assign `player_ship` sprite
 - **BoxCollider2D** → Is Trigger ✓, Size (0.5, 0.6)
 - **Rigidbody2D** → Gravity Scale: 0, Freeze Rotation ✓
 - **HealthSystem** script → Max Health: 5
 - **PlayerController** script
3. Set tag to **Player**
4. Create child "FirePoint" at position (0, 0.5, 0)
5. On PlayerController: assign bullet prefab and fire point
6. Drag to `Assets/Prefabs/Player/`

Enemy Prefabs

Repeat for each enemy type (Basic, Fast, Tank, Boss):

1. Create empty GameObject with enemy name
2. Add: SpriteRenderer, BoxCollider2D (trigger), Rigidbody2D (no gravity)
3. Add the appropriate enemy script (BasicEnemy, FastEnemy, etc.)
4. Add **BulletPattern** component, assign enemy bullet prefab
5. Set tag to **Enemy**
6. Drag to `Assets/Prefabs/Enemies/`

Recommended values:

Enemy	Health	Speed	Score	Fire Rate
Basic	3	3.0	100	1.5s
Fast	2	5.0	150	2.0s
Tank	8	1.5	300	1.0s
Boss	50	2.0	5000	0.8s

Bullet Prefabs

For each bullet type:

1. Create GameObject with SpriteRenderer and appropriate bullet sprite
2. Add BoxCollider2D (trigger) and Rigidbody2D (no gravity)
3. Add **Bullet** script
4. Set tag to `PlayerBullet` or `EnemyBullet`
5. Save as prefab

Power-Up Prefabs

For each power-up type (Health, Shield, RapidFire, SpreadShot, ExtraLife, ScoreBonus):

1. Create GameObject with SpriteRenderer and matching sprite
2. Add CircleCollider2D (trigger, radius 0.3) and Rigidbody2D (no gravity)
3. Add **PowerUp** script, set the Type enum
4. Set tag to `PowerUp`
5. Save as prefab

Create Scenes

MainMenu Scene

1. **File → New Scene** (Basic 2D)
2. Set Camera background to dark blue (0.02, 0.02, 0.08)
3. Add **StarfieldGenerator** to an empty GameObject
4. Create Canvas (Screen Space - Overlay):
 - Title Text: "SPACE SHOOTER" (font size 48, cyan)
 - High Score Text
 - Start Button
 - Quit Button
 - Music/SFX volume sliders
5. Add **MainMenuController** script to a GameObject
6. Wire button onClick events to the controller
7. Add GameManager, SoundManager, InputHandler GameObjects
8. **Save as** `Assets/Scenes/MainMenu.unity`

GamePlay Scene

1. **File → New Scene** (Basic 2D)
2. Camera: orthographic, size 5, dark background
3. Add background layers with **ParallaxBackground**
4. Add **StarfieldGenerator**
5. Place **Player** prefab at (0, -3.5, 0)

6. Add **EnemySpawner** GameObject:
 - Assign enemy prefabs to the serialized fields
7. Create HUD Canvas with:
 - Score text (top-left)
 - Lives text + icons
 - Health bar (Slider)
 - Wave announcement text (center, CanvasGroup for fade)
 - Pause panel (hidden by default)
8. Add **HUDController** and wire all UI references
9. **Save as** `Assets/Scenes/GamePlay.unity`

GameOver Scene

1. Similar to MainMenu but with:
 - “GAME OVER” title in red
 - Final score and high score text
 - “NEW HIGH SCORE!” conditional text
 - Restart, Main Menu, and Quit buttons
2. Add **GameOverController** and wire references
3. **Save as** `Assets/Scenes/GameOver.unity`

Wire Up References

GameManager (on MainMenu scene)

- Assign all 6 power-up prefabs to the `powerUpPrefabs` array

PlayerController

- Bullet Prefab → PlayerBullet prefab
- Fire Point → child FirePoint transform
- Shield Visual → ShieldVisual prefab (or child object)
- Explosion Prefab → Explosion prefab

Enemy Scripts

- Each enemy’s BulletPattern → assign EnemyBullet prefab (or BossBullet for boss)
- Explosion Prefab → Explosion prefab

EnemySpawner

- Basic/Fast/Tank/Boss enemy prefab references

HUDController

- Wire all Text, Slider, Image, and panel references

MainMenuController

- Wire button and slider references

GameOverController

- Wire all text and button references

Configure Tags

Ensure these tags exist (**Edit** → **Project Settings** → **Tags and Layers**):

Tag	Used By
Player	Player ship
Enemy	All enemy ships
PlayerBullet	Player's projectiles
EnemyBullet	Enemy projectiles
PowerUp	All power-up pickups

Testing in Editor

1. Open `Assets/Scenes/MainMenu.unity`
2. Press **Play** 
3. Click **START GAME**
4. **Controls:**
 - **WASD** or **Arrow Keys** — Move
 - **Space** or **Left Mouse** — Fire (also auto-fires)
 - **Escape** — Pause/Resume
5. Verify:
 -  Player moves within screen bounds
 -  Bullets fire and destroy enemies
 -  Enemies spawn in waves with increasing difficulty
 -  Score updates on enemy kills
 -  Power-ups drop and apply effects
 -  Health bar updates on damage
 -  Lives decrement on death with respawn
 -  Game over triggers at 0 lives
 -  Pause menu works
 -  Scene transitions work (menu → game → game over → menu)

Build for Windows

Configure Build Settings

1. **File** → **Build Settings** (Ctrl+Shift+B)
2. Verify these scenes are listed in order:
 - `Scenes/MainMenu` — Index 0
 - `Scenes/GamePlay` — Index 1
 - `Scenes/GameOver` — Index 2
3. If missing, click **Add Open Scenes** for each
4. Set **Platform** to **Windows, Mac, Linux** (PC)
5. Click **Switch Platform** if needed

Configure Player Settings

1. Click **Player Settings** (bottom-left of Build Settings)
2. Set:
 - **Company Name:** Your name/studio
 - **Product Name:** Space Shooter
 - **Default Icon:** (optional, can use player ship sprite)
 - **Resolution:**
 - Default Width: 800
 - Default Height: 600
 - Fullscreen Mode: Windowed (or Fullscreen Window)
 - Resizable Window: ✓
 - **Splash Screen:** Customize or disable (requires Unity Pro)
3. Under **Other Settings:**
 - **Scripting Backend:** Mono (faster builds) or IL2CPP (better performance)
 - **Api Compatibility Level:** .NET Framework or .NET Standard 2.1

Build

1. In Build Settings, click **Build**
2. Choose an output folder (e.g., `Builds/Windows/`)
3. Name the executable: `SpaceShooter.exe`
4. Click **Save**
5. Wait for the build to complete
6. The output folder will contain:

```
Builds/Windows/
├─ SpaceShooter.exe           ← Run this!
├─ SpaceShooter_Data/        ← Game data (required)
├─ UnityCrashHandler64.exe    ← Crash handler
└─ UnityPlayer.dll            ← Unity runtime
```

Build & Run

- Alternatively, click **Build And Run** to build and launch immediately

Distribution

To distribute the game, **zip the entire output folder** including:

- The `.exe` file
- The `_Data` folder
- All `.dll` files

All files must stay together in the same directory.

Troubleshooting

Scripts Won't Compile

- Ensure you're using Unity 2021.3+ with 2D packages
- Check **Window** → **Console** for exact error messages
- Verify all script files are in the correct folders under `Assets/Scripts/`
- Ensure namespace references match (`SpaceShooter.Player`, `SpaceShooter.Enemy`, etc.)

Tags Not Found

- Open **Edit → Project Settings → Tags and Layers**
- Add missing tags: `Player` , `Enemy` , `PlayerBullet` , `EnemyBullet` , `PowerUp`

Missing References on Prefabs

- Select each prefab and check the Inspector for yellow warning icons
- Re-assign any missing sprite, prefab, or component references
- Use the Game Setup Wizard to recreate prefabs if needed

No Enemies Spawning

- Verify EnemySpawner has enemy prefab references assigned
- Check that enemy prefabs have the “Enemy” tag
- Ensure `GameManager.StartGame()` is being called (check main menu button)

Player Not Taking Damage

- Verify enemy bullets have tag “EnemyBullet”
- Check that colliders are set to “Is Trigger”
- Verify both objects have `Rigidbody2D` components

No Sound

- `SoundManager` needs `AudioClip` references assigned
- The game works without sound — audio is optional
- You can add `.wav` or `.ogg` files to `Assets/Audio/SFX/` and assign them

Build Fails

- Check Console for build errors
 - Ensure all scenes are added to Build Settings
 - Try **Clean Build** (delete the output folder first)
 - Switch scripting backend to Mono if IL2CPP fails
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Project Structure Reference

space_shooter_game/	
Assets/	
Scripts/	
Player/	
PlayerController.cs	Player movement, shooting, power-ups
HealthSystem.cs	Reusable HP component
Enemy/	
EnemyBase.cs	Base enemy class
BasicEnemy.cs	Straight-moving enemy
FastEnemy.cs	Zigzag movement enemy
TankEnemy.cs	Heavy, slow enemy
BossEnemy.cs	Multi-phase boss
EnemySpawner.cs	Wave-based spawn system
Weapons/	
Bullet.cs	Basic projectile
HomingBullet.cs	Target-tracking projectile
BulletPattern.cs	Pattern firing (spread, circle, aimed)
PowerUps/	
PowerUp.cs	All power-up types
Managers/	
GameManager.cs	State, score, lives, scenes
SoundManager.cs	Audio playback
InputHandler.cs	Centralized input
ScoreManager.cs	Score + combo system
UI/	
HUDController.cs	In-game HUD
MainMenuController.cs	Main menu logic
GameOverController.cs	Game over screen
Effects/	
ParallaxBackground.cs	Scrolling backgrounds
ExplosionEffect.cs	Explosion animation
StarfieldGenerator.cs	Particle starfield
Utils/	
ObjectPooler.cs	Object pooling
AutoDestroy.cs	Timed self -destruct
ScreenWrapper.cs	Screen edge wrapping
Editor/	
ProceduralSpriteGenerator.cs	Sprite creation tool
GameSetupWizard.cs	One-click setup wizard
Sprites/	Generated sprite PNGs
Prefabs/	Game object prefabs
Scenes/	
MainMenu.unity	
GamePlay.unity	
GameOver.unity	
Audio/	
Music/	Background music (add .ogg/.wav)
SFX/	Sound effects (add .ogg/.wav)
Resources/	
ProjectSettings/	Unity project configuration
Packages/	
manifest.json	Package dependencies
BUILD_INSTRUCTIONS.md	This file

Game Features Summary

Feature	Implementation
Player Movement	WASD/Arrows, clamped to screen bounds
Shooting	Auto-fire + Space/Mouse, configurable rate
Enemy Types	Basic, Fast (zigzag), Tank (pause+fire), Boss (phases)
Wave System	Predefined + procedural waves, boss every 5th
Bullet Patterns	Straight, Spread3, Spread5, Circle, Aimed, Homing
Power-Ups	Health, Shield, Rapid Fire, Spread Shot, Extra Life, Score
Health System	Player HP bar, enemy HP, damage events
Scoring	Points per kill, high score persistence (Player-Prefs)
Parallax BG	Multi-layer scrolling + particle starfield
Sound	SFX pool system, music per scene, volume controls
UI	HUD, wave announcements, pause menu
Scenes	Main Menu → Gameplay → Game Over (full flow)
Persistence	High score saved between sessions

Created with ❤️ — A complete, production-ready Unity space shooter game.